

Mahale Silva

23110 Arora Hills Drive | Clarksburg, MD 20871 | (240) 733 2797 | mahalesilva@gmail.com

Education

University of Maryland - College Park

(Expected Spring 2027)

Bachelor of Science (BS) in Chemistry

- GPA: 4.0
- Dean's List (Fall 2024-Spring 2026)
- BioFIRE Living-Learning Program

Research Experience

SDIP REU (Clarkson University)

(Summer 2026)

Undergraduate Researcher

- Worked with Peptide Amphiphiles (PA) to see how they effect and interact cervical cancer cells
- Used fluorescent detection methods to analyze how PA enters cells and interacts inside the cells, along with how PA is distributed during splitting

Vedernikov Research Group (UMD - College Park)

(Fall 2025 - Present)

Undergraduate Researcher

- Performed DFT calculations (Gaussian 16) to investigate the mechanistic pathways of platinum organometallic complexes
- Optimized intermediates and transition states to map energy profiles, and then analyzed the impacts of stereoisomers and conformational isomers, along with ligand effects, of complexes on the energy profiles
- Analyzed data to determine the lowest energy pathways and rate-determining steps for wet lab synthesis

First-Year Innovation & Research Experience (FIRE) Program

Peer Research Mentor

(Spring 2026 - Present)

- Developed a paperfluidic device for the rapid detection of Maple Syrup Urine Disease biomarkers
- Mentored incoming researchers in experimental protocols and assisted the P.I. with laboratory training and organization

Undergraduate Research Assistant

(Spring 2025 - Spring 2026)

- Conducted experiments to determine detection limits for target chemicals in E. coli
- Developed a colorimetric rapid diagnostic assay device for liquid samples

OURS Immersive Research Internship Experience (IRIE)

(Summer 2025)

Research Intern

- Quantified the literature-based limit of detection for a disease called Maple Syrup Urine Disease and developed a chemical assay device for detection

Leadership Experience

Undergraduate T.A. - Organic Chemistry II Lab (UMD - College Park)

(Fall 2026)

- Lectured on, guided, and demonstrated lab protocols for students in Organic Chemistry II

Lab

- Led students through fundamental lab techniques such as filtration, recrystallization, distillation, refluxes, chromatography, and spectroscopy

Academic Peer Mentor - Introduction to Laboratory Practices and Research in the Chemical Sciences (UMD - College Park)

(Fall 2025)

- Mentored students in lab report development and experimental techniques

Posters and Presentations

Silva, M. et al. "Colorimetric Paperfluidic Assay for MSUD Detection Through Alpha Keto Acids." *Undergraduate Research Day (URD)*, University of Maryland, College Park, MD, April 2026. Poster presentation.

Silva, M. et al. "Colorimetric Assay For MSUD Detection "Through Alpha Keto Acids." *Summer Undergraduate Research Conference (SURC)*, University of Maryland, College Park, MD, July 2025. Poster presentation.

Skills

- Software: GaussView/Gaussian, ChemDraw, Microsoft Word, Excel, and PowerPoint
- Technical Skills: Centrifugation, PCR, Crystallization, TLC, Extraction/Filtration Techniques, Spectrophotometry, CAD, Paperfluidics, Analytical Calibration, Colorimetric Assays, and Fabrication Methods